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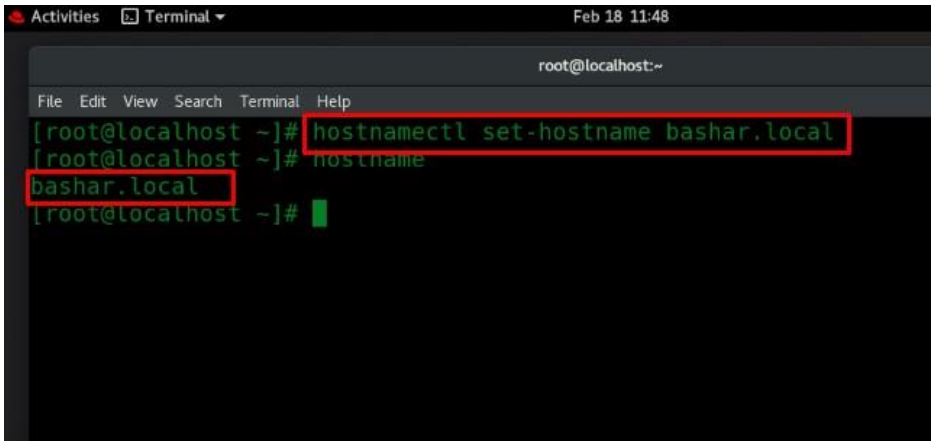
Project Overview

This project involved configuring a Red Hat Enterprise Linux 8 (RHEL 8) system for a corporate environment named **SERVOTECH Incorporation**. The primary objective was to establish a structured network identity, manage departmental user access, and implement a secure file management system.

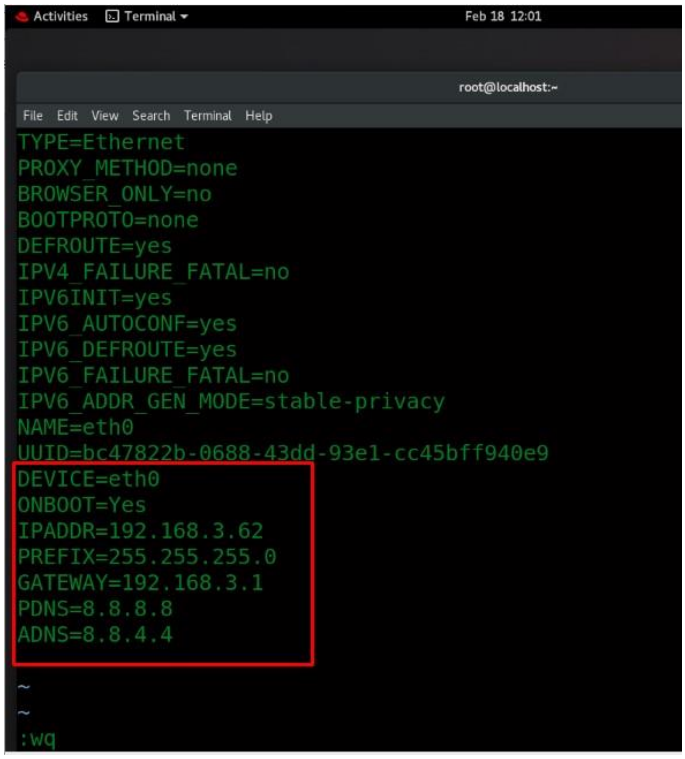
1. System Settings

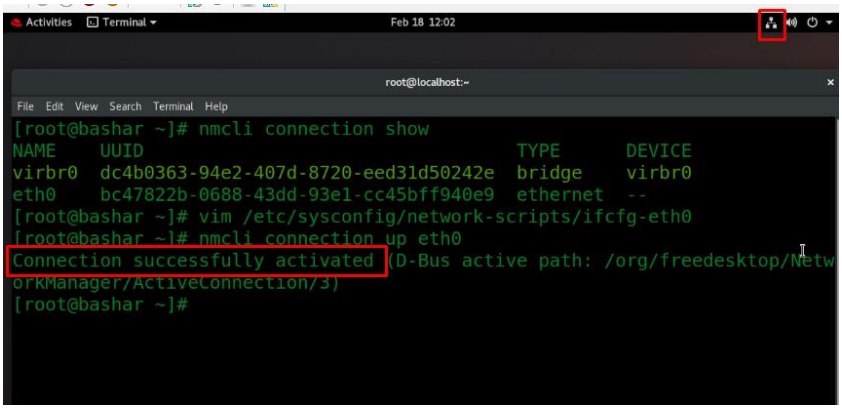
This step was performed to ensure the server is identifiable and maintains a stable connection within the corporate network.

- **Hostname Configuration:** The hostname was set to bashar.local using the command "**hostnamectl set-hostname**" bashar.local to reflect the server's specific role.



- **Static IP Setup:** To prevent connectivity issues, a static IP was configured by editing the network script /etc/sysconfig/network-scripts/ifcfg-eth0.
- **IP Address:** 192.168.3.62
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.3.1 ○
- **DNS:** Primary (8.8.8.8) and Alternate (8.8.4.4).





- **Activation:** The connection was successfully activated using the **nmcli connection up eth0** command.

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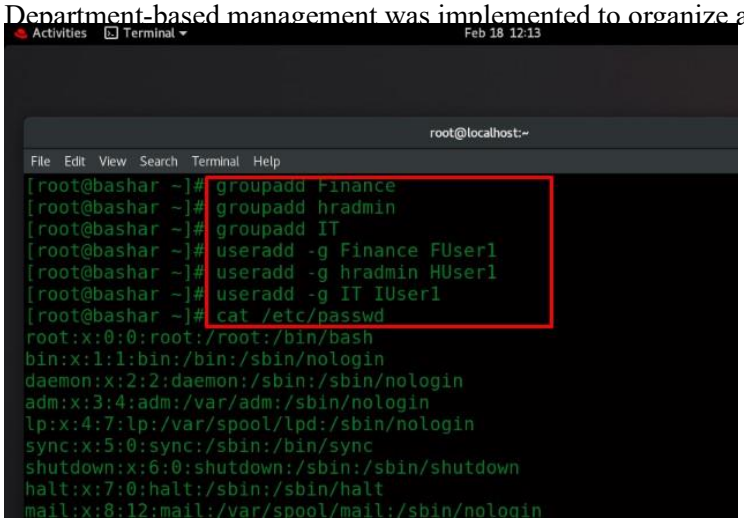
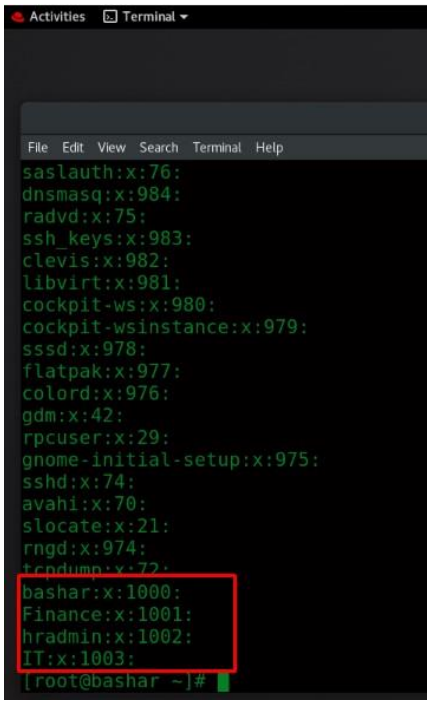
2. User and Group Management

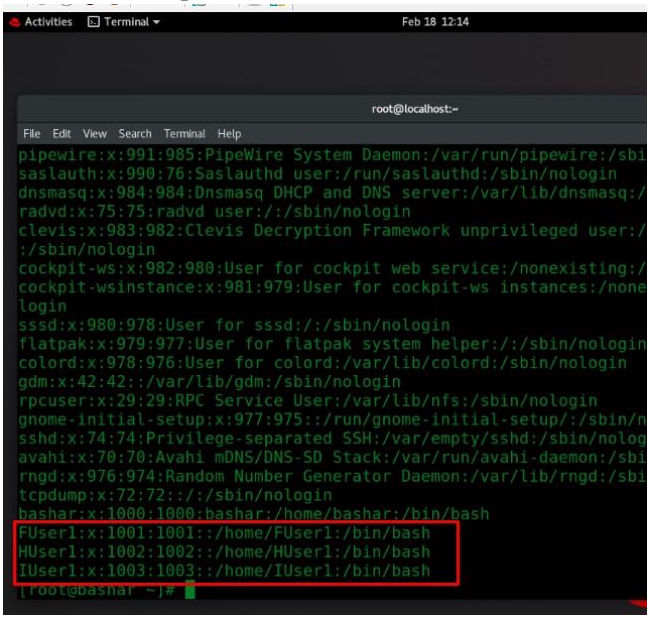
Department-based management was implemented to organize access and administrative control.



- **Groups:** Created three primary groups: Finance, hradmin, and IT.



- **Users:** Created at least one user per group (e.g., FUser1, HUser1, and IUser1) to represent the different departments.



- **Purpose:** This ensures that employees are categorized correctly for security and permission auditing.

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3. Directory Structure and Permissions

A centralized data structure was built to secure departmental files.

- **Directories:** Created a parent directory /**orgdata** with subdirectories for Finance, hradmin, and IT.

```
File  Edit  View  Search  Terminal  Help
[root@bashar ~]# mkdir -p /orgdata/Finance /orgdata/hradmin /orgdata/IT
[root@bashar ~]# ls
anaconda-ks.cfg  Documents  initial-setup-ks.cfg  Pictures  Templates
Desktop          Downloads  Music                  Public    Videos
[root@bashar ~]# cd /r
root/ run/
[root@bashar ~]# ls -r /orgdata/
IT  hradmin  Finance
[root@bashar ~]#
```

- **Permissions & ACLs:** * Used **chmod 770** to ensure that only the owner and group members can access departmental folders.

```
Activities  Terminal ▾                    Feb 18 13:10
                          IUser1@localhost:/root
File  Edit  View  Search  Terminal  Help
[root@bashar ~]# setfacl -m g:IT:rwx /orgdata/Finance/
[root@bashar ~]# setfacl -m g:IT:rwx /orgdata/hradmin/
[root@bashar ~]# setfacl -m g:IT:rwx /orgdata/IT/
[root@bashar ~]# setfacl -m g:Finance:rwx /orgdata/Finance/
[root@bashar ~]# setfacl -m g:hradmin:rwx /orgdata/hradmin/
[root@bashar ~]# chmod 770 /orgdata/Finance/
[root@bashar ~]# chmod 770 /orgdata/hradmin/
[root@bashar ~]# chmod 770 /orgdata/IT/
[root@bashar ~]# getfacl /orgdata/Finance/
getfacl: Removing leading '/' from absolute path names
# file: orgdata/Finance/
# owner: root
# group: root
user::rwx
group::rwx
group:Finance:rwx
group:IT:rwx
mask::rwx
other::---

[root@bashar ~]# su IUser1
[IUser1@bashar root]$ ls /orgdata/Finance/
[IUser1@bashar root]$ ls /orgdata/hradmin/
[IUser1@bashar root]$ su FUser1
```

```
Activities  Terminal ▾                    Feb 18 12:43
                          root@localhost:~
File  Edit  View  Search  Terminal  Help
[root@bashar ~]# getfacl /orgdata/hradmin/
getfacl: Removing leading '/' from absolute path names
# file: orgdata/hradmin/
# owner: root
# group: root
user::rwx
group::rwx
group:Finance:rwx
group:IT:rwx
mask::rwx
other::---

[root@bashar ~]# getfacl /orgdata/IT/
getfacl: Removing leading '/' from absolute path names
# file: orgdata/IT/
# owner: root
# group: root
user::rwx
group::rwx
group:IT:rwx
mask::rwx
other::---

[root@bashar ~]# clear
```

- **IT Access:** Applied **Access Control Lists (ACLs)** using **setfacl** to grant the IT user (IUser1) full access to all directories

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```
Activities  Terminal ▾                    Feb 18 13:17
                    FUser1@localhost:/root
File  Edit  View  Search  Terminal  Help
[root@bashar ~]# setfacl -m g:hradmin:rwx /orgdata/hradmin/
[root@bashar ~]# chmod 770 /orgdata/Finance/
[root@bashar ~]# chmod 770 /orgdata/hradmin/
[root@bashar ~]# chmod 770 /orgdata/IT/
[root@bashar ~]# getfacl /orgdata/Finance/
getfacl: Removing leading '/' from absolute path names
# file: orgdata/Finance/
# owner: root
# group: root
user::rwx
group::rwx
group:Finance:rwx
group:IT:rwx
mask::rwx
other::---

[root@bashar ~]# su IUser1
[IUser1@bashar root]$ ls /orgdata/Finance/
[IUser1@bashar root]$ ls /orgdata/hradmin/
```

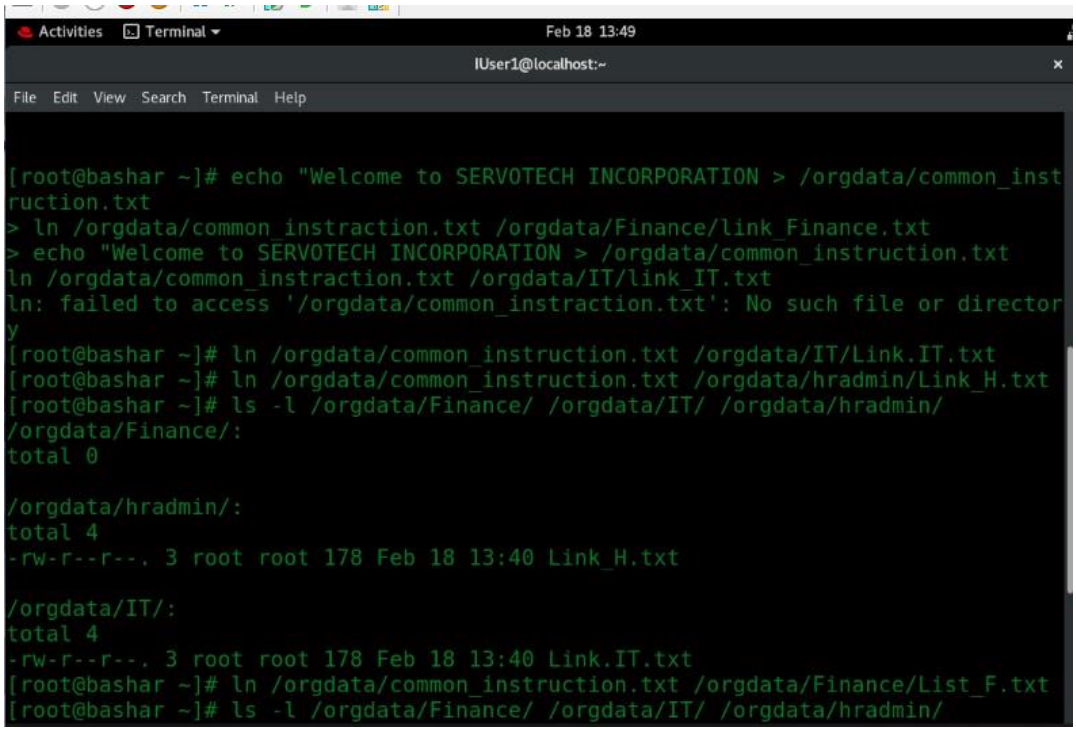
- **Restriction:** Verified that other users (like FUser1 or HUser1) are strictly restricted to their own directories and denied access elsewhere.

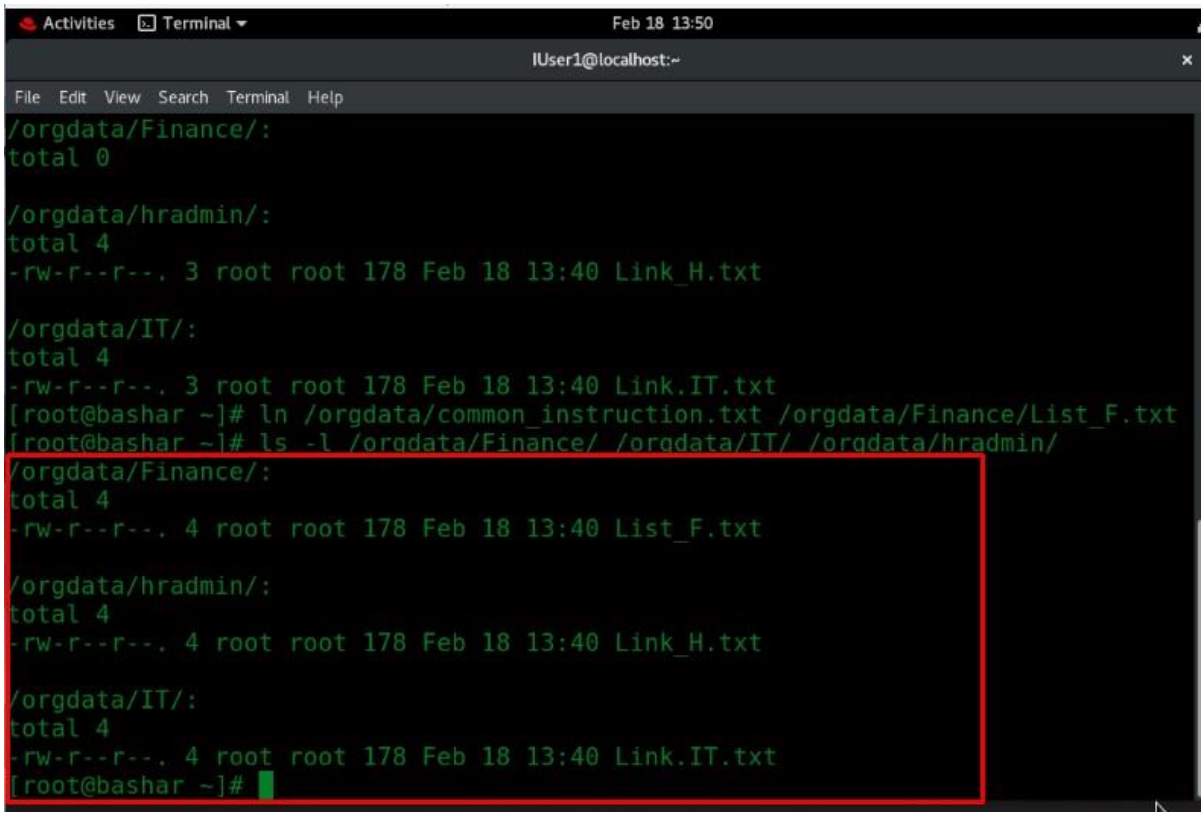
```
Activities  Terminal ▾                    Feb 18 13:23
                    IUser1@localhost:/root
File  Edit  View  Search  Terminal  Help
[FUser1@bashar root]$ ls /orgdata/hradmin/
ls: cannot open directory '/orgdata/hradmin/': Permission denied
[FUser1@bashar root]$ ls /orgdata/IT/
ls: cannot open directory '/orgdata/IT/': Permission denied
[FUser1@bashar root]$ ls /orgdata/Finance/
[FUser1@bashar root]$ su
Password:
[root@bashar ~]# su HUser1
[HUser1@bashar root]$ ls /orgdata/hradmin/
[HUser1@bashar root]$ ls /orgdata/Finance/
ls: cannot open directory '/orgdata/Finance/': Permission denied
[HUser1@bashar root]$ ls /orgdata/IT
ls: cannot open directory '/orgdata/IT': Permission denied
[HUser1@bashar root]$ su
Password:
[root@bashar ~]# su IUser1
[IUser1@bashar root]$ ls /orgdata/IT/
[IUser1@bashar root]$ ls /orgdata/hradmin/
[IUser1@bashar root]$ ls /orgdata/Finance/
```

4. Hard Link Implementation

This was used to share a single source of information across multiple departments efficiently.

- **Central File:** Created a master file at /orgdata/common_instructions.txt.
- **Linking:** Used the ln command to create hard links within each department folder (e.g., List_F.txt, Link_H.txt, and Link_IT.txt).





- **Demonstration:** This setup proves that editing the content in any of these locations updates all links simultaneously because they all point to the same data on the disk.

Conclusion: The system is now fully configured with a professional network identity, departmental user hierarchy, and a secure, linked file system.